Kinect Lite: Results (depth-only ICP)

Two depth cleaners are compared: our **9×9 median** and the **paper** (Fu et al., 2012). Each cleaned sequence is tracked with depth-only ICP and the camera path is compared with the true path. ATE is total drift (cm), RPE the error per step (mm); lower is better. Datasets: ICL-NUIM living room kt0 (150 tracked frames) and TUM RGB-D fr3 long_office_household (150 tracked frames).

1. Depth accuracy (ICL, against the true depth)

Method	MAE (mm)	RMSE (mm)	Median err (mm)	Within 1 cm
No cleaning	39.2	187.5	13.3	42.2%
Median 9×9	**18.8**	**55.4**	**12.3**	**44.7%**
Paper	37.7	186.0	13.4	33.7%

Source: `src/results/icl_compare.txt`

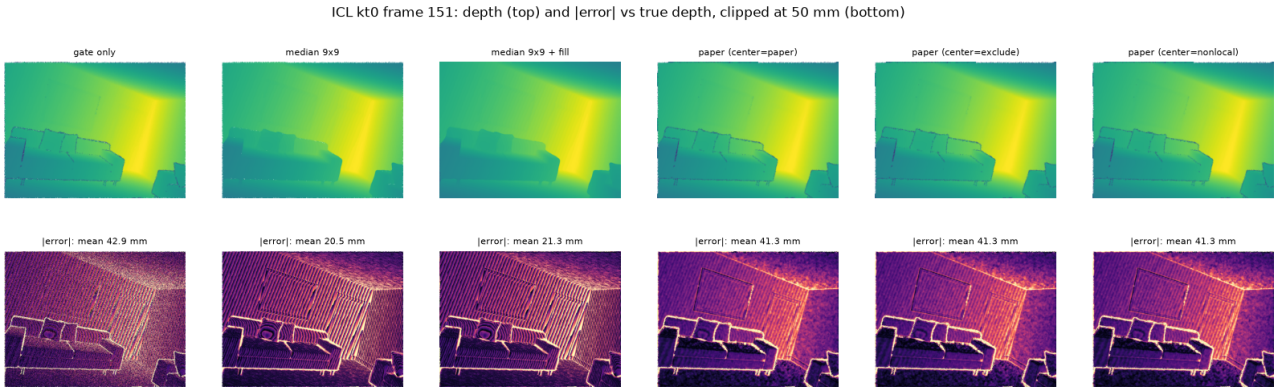


Figure 1: ICL frame 151: cleaned depth (top) and error against the true depth (bottom).

2. Camera tracking with point-to-plane ICP

Method (ATE / RPE)	ICL	TUM
True depth	0.19 / 0.07	–
No cleaning	17.07 / 14.43	13.97 / 11.69
Median 9×9	14.94 / 11.78	13.32 / 10.86
Paper	**0.95 / 2.02**	**3.33 / 4.37**

Source: `src/results/icl_compare.txt, tum_compare.txt`

The paper has the worse per-pixel depth error but tracks about 6× better per step.

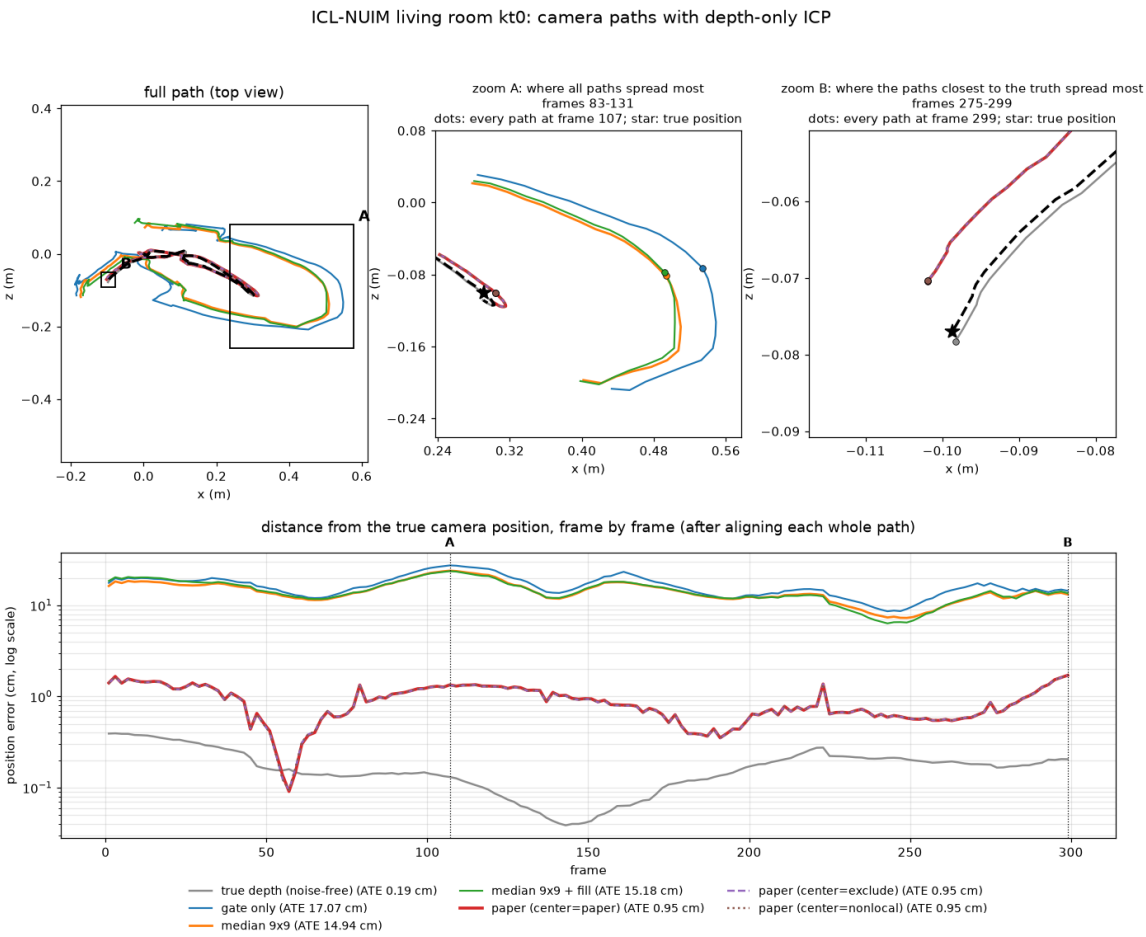


Figure 2: ICL: estimated camera paths (top) and position error per frame (bottom).

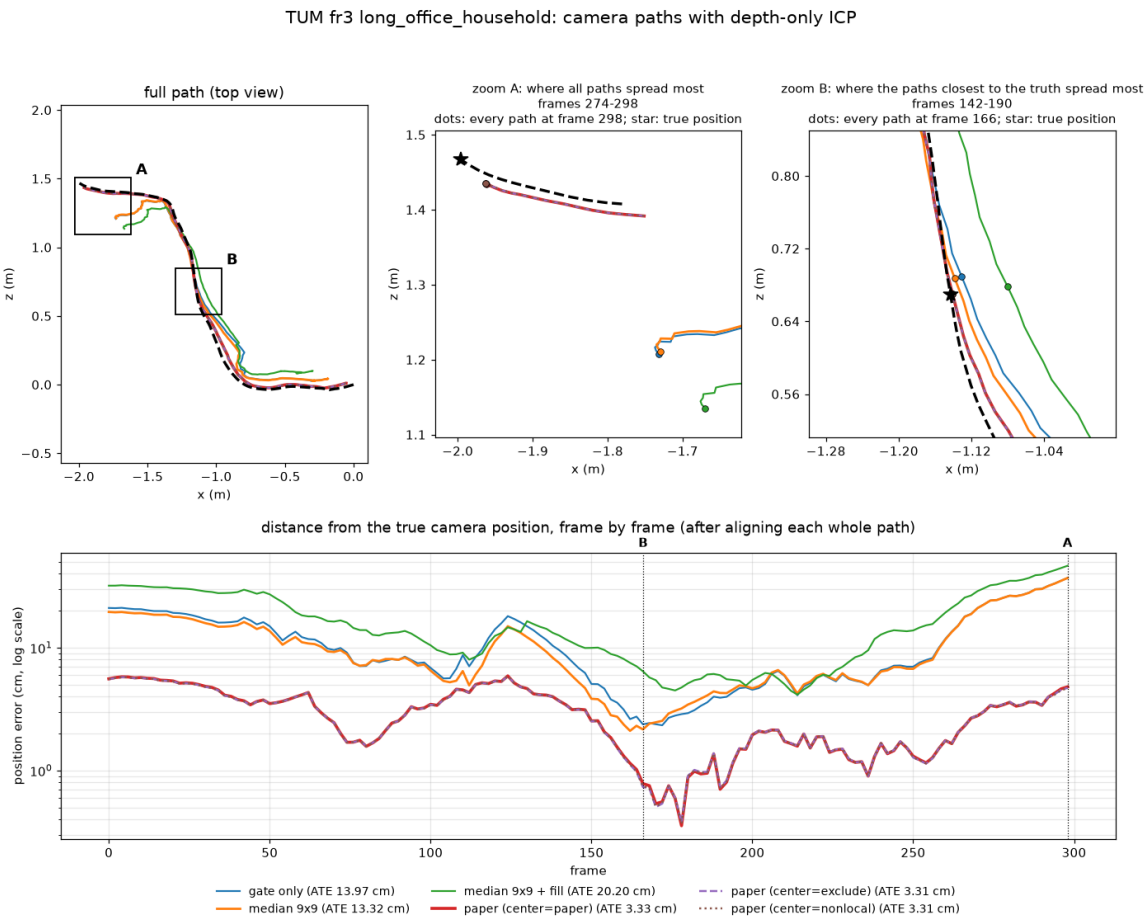


Figure 3: TUM: estimated camera paths (top) and position error per frame (bottom).

3. ICP variants (ATE / RPE)

	Depth	Point-to-point	Point-to-plane	Plane + Tukey	GICP
ICL	True depth	1.81 / 3.29	0.19 / 0.07	0.14 / 0.05	–
	No cleaning	36.96 / 21.36	17.05 / 14.38	22.47 / 17.47	–
	Median	28.50 / 19.72	14.94 / 11.78	19.24 / 14.11	–
	Paper	1.93 / 3.16	0.95 / 2.02	1.42 / 2.54	–
TUM	No cleaning	27.24 / 14.92	13.97 / 11.69	14.31 / 11.55	–
	Median	18.14 / 13.78	13.33 / 10.86	13.07 / 10.96	diverged
	Paper	5.31 / 5.21	3.33 / 4.37	3.21 / 4.23	14.13 / 43.42

Source: `src/results/icp_variants.txt`

Point-to-plane beats point-to-point in all seven rows. The robust Tukey loss gives no clear gain. The paper beats the median under every variant that ran. Generalized ICP produced non-finite poses on noisy depth (no ICL numbers; TUM median diverged) and was not tuned. The numbers come from the log of a Kaggle run that crashed at the final table step, after all other tracking had finished.

4. Why the median does not help ICP (9 ICL frames)

	Normal error vs true (deg)		Consecutive-frame disagreement (mm)	
	median	mean	median $\lvert r \rvert$	bias
True depth	–	–	0.3	0.01
No cleaning	5.5	10.1	5.0	0.84
Median	2.8	7.4	3.6	1.09
Paper	**1.6**	**3.7**	**1.8**	**0.26**

Source: `experiments/normals_check/results/normals.txt`

The paper's normals are about twice as accurate and its consecutive frames agree about twice as well. The check uses 9 frames on ICL only and cannot separate smoothing from temporal averaging.

5. Hole filling (ATE in cm, depth-only ICP)

Fill method	Fill err (mm)	ICL	TUM	Time/frame
None (keep holes)	–	**14.94**	13.33	0 ms
Telea inpainting	87.0	15.18	20.21	210 ms
Navier–Stokes	22.5	15.36	**12.03**	224 ms
Nearest pixel	22.3	15.05	17.54	50 ms
Region growing (small holes)	15.5	14.94	13.32	52 ms
Region growing (median)	22.5	15.15	15.14	386 ms
Background	83.1	15.24	13.51	40 ms
Temporal (paper)	**12.2**	15.11	13.37	7.9 s

Source: `experiments/hole_filling/results/fills.txt`

No fill reliably improves tracking. The one gain (Navier–Stokes on TUM) does not hold on ICL.

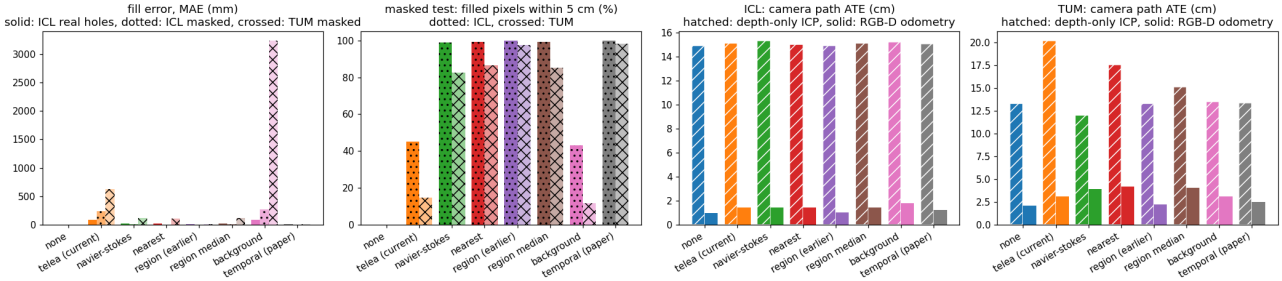


Figure 4: Hole-filling comparison. The tracking panels in this figure were produced by an earlier version of the script that also showed another tracker; the depth-only ICP values are those in the table above.